

Introduction To Body Sensor Networks

Ritu Singh¹, Mansi V Gaonkar¹, Sakshi Sharma¹, Priyanka Grover¹, Aanchal Khatri²

¹Department of Computer science & Engineering

Manav Rachna International Institute of Research And Studies,

²Accendere KMS Ltd.

Faridabad, India

ritun22101998@gmail.com, mvgaonkar1999@gmail.com, saku5433@gmail.com,
priyankagrover.fet@mriu.edu.in, aanchal.khatri@accendere.co.in

Abstract— As we all know that in the health sector the use/utilization of information technology is less especially in areas of operational efficiency due to which there is limiting collaboration and coordination between patient and physician relationship. So to solve these, cloud has the ability to overcome this type of disadvantages. For instance one of the best method are the Healthcare systems that works according to the cloud and its services which provide us with various range of advantages such as economical and the functionality of service is also very good as compared to that of in house clients. As the storing content of the healthcare services is maximum the user or any other health provider can easily store large bytes of data such as X-ray, sonography image and the other images which are being commonly used in the department of radiology by the doctor. Many users especially the health experts and organizations opt for the private cloud due to the sensitivity of the data. Further in this paper we would be telling you about the body sensor network and how it is capable of maintaining the user life. Through this paper we would be trying to give you an idea that how these sensors can be used by the patients especially the elderly people or those who have the disease known as Alzheimer. Alzheimer refers to the illness in which the patient suffers from permanent or temporary memory loss and fails to recognize and face difficulty in decision making.

Keywords - Cloud Healthcare, Sensors, BSN, Body Sensor, Cloud computing

I. INTRODUCTION

As the life expectancy level of population is increasing, many developed countries have been left with an ageing population needing medical care at a time where resources such as healthcare professionals, community care and financial help are being stretched in order to meet the increased demand due to which various health providers look for more innovative and cost-effective solutions to address this growing problem. [1] Cloud Healthcare is one such solution of it. It is a cloud service used for the storing of personal health information. It helps the patient get smarter healthcare decisions, solutions and mainly focus on the patient relationship towards doctor and his health.

To solve this type problem, IBM and Active Health Management, a subsidiary of a leading American provider of healthcare service in November, 2010 developed a new clinical information management system, called a "Collaborative Care Solution" based on cloud computing architecture. Secondly, various online health recording systems are made so that the life of people get healthy and more active. For instance, Electronic medical record is one

such example in which the patient records are stored in cloud so that in future whenever and wherever the patient requires the document it would be easily available.

In this paper our main work is to highlight the importance of body sensor networks so that people who need the most of it can use the sensor and make their life peaceful.

This paper mainly describe how cloud can be helpful in providing more health information and maintaining the health records so that there is a decrease in health-related problems all over the world. The whole paper is structured as following. Section II consist of the overview of Cloud Healthcare and its related work. Section III discusses about our proposed framework. Section IV consist of overview of body sensor networks (BSN). Section V consist of working of sensors. Section VI discusses about importance of BSN in today's scenario. Section VII provides conclusion works.

Cloud Healthcare is a service through which the personal health information of the patient can be stored, maintained. In this system, a backup is present in such a way that if due to some reasons our records get lost or destroyed we can easily retrieve or get it back. In today's world various health organizations are choosing private cloud services rather than public services due to the presence of sensitive data in it. In this type of system, there are various advantages. Firstly, the solution aim to enable easy access by any medical and healthcare service to a wide range of information from various sources such as electronic medical records (EMRs), medication and laboratory data. [2] This type of solution is also able to highlight trends that how patients are responding well to the treatment. Secondly it helps the user by providing and analyzing the performance of our health in an advanced manner in such a way that the physician or the entire healthcare organization work on the report against the national or hospital quality standards. Thirdly and one of the most important thing is that the users are charged a fixed monthly fee without having to make significant upfront investments thus overcoming the challenges of updating system when clinical requirement or any reporting guideline change.

II. PROPOSED FRAMEWORK

Through this paper, our main framework is to create a device or instrument which would help the patient especially the elderly people and the alzheimer patient to make quick decision and maintain their dietary balance. Secondly it would help their family members to keep the track of them and if by

chance if there is a change in their body conditions then would be immediately noted about it along with their location. In this device we are planning to insert a sensor which can be easily attached to the brain cells of the patient and help them to make their decision and recognition capability better. Secondly it would also help them to maintain their diet by making them remind of their schedule. Third and most important we would attach a GPS system in it, so that their family members can easily locate them. Finally using the cloud, the patient can maintain their medical data or record in their mobile or laptop. Our main aim is to make the life of people more enjoyable as well as healthy. In other word we can also say that using this device we are trying to make them have better decisions.

III. BODY SENSOR NETWORKS

Body Sensor Network is basically a wireless sensor network. This network consists of sensor nodes which help in monitoring the vital condition of our body such as the physical and human condition. It is mostly used in medical field. The main work of this sensors is to check the status of the patient condition through signals which are remotely accessed. As this sensor provide large amount of real-time data, cloud is used so that details can be send and saved in server and transferred to healthcare organization where data is being analyzed and interpreted by medical staff to get the details of patient from which illness or disease he/her is suffering.[3] Cloud offers the way or we can say technique to utilize various sensors, storage process, analyze and share results with EHRs. Body cloud is basically a link between body networks and cloud computing which allows collection and transferring of data to various health organizations and medical websites. The functionality of sensor works in two ways- either sensor is attached to the body surface or implanted in body without causing harm to patient every day normal activities. BSN applications are mostly cost effective.

Their main aim is to make the life of patient very healthy and maintain a balanced diet. Let take the example of pacemaker. Pacemaker is also a body sensor through which the unconditional heart beat is checked. One such example of wireless network is wireless capsule. This capsule main work is to record the images of intestinal activities and tracts.

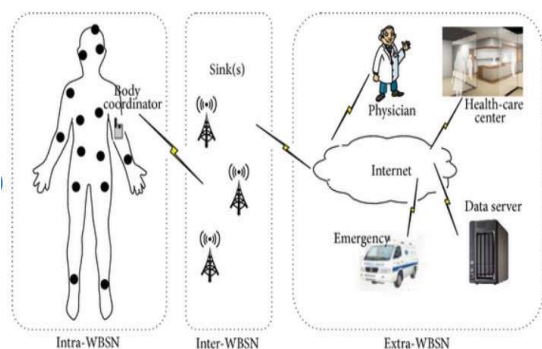


Fig 1. Working of WBSN's

A. Parameters on basis of which BSN works

1. **Interference:** The user must keep less communication between the sensor nodes implanted [4] inside or outside the body to check temperature, heat radiation or by other wireless network in such a way we obtain reliable transmission of signals and we obtain the correct details of our body.
2. **Transmission range:** The user must always keep in the mind that distance for transmission must have range of 2-5 meters for better transmission of signals between the sensor nodes.
3. **Controlling Error:** For increasing the rate and the connectivity of network the data which is analyzed and recorded should not be pirated, lost. For the wave propagation in the body there must be low path losses.
4. **Security:** To prevent the network from any kind of disturbance such as jamming, co-interference among nodes, tampering, spoofing or so on, the network preferred must be secured as well as authenticated.
5. **Service Quality:** In this the user check how, much time would the device or the sensor would work, would check if it easy to use and does it have any disadvantage or side effect on the body. QoS maintenance is necessary for performance management.
6. **Network density:** For better network, the connectivity must be high so that no problem is caused to the patient during the transmits of signals from patient body to the computer.
7. **Scalability:** There must be at least 15-20 nodes attached according to the system so that the sensor works efficiently without any problem or stoppage.
8. **Data rates:** One of the most supported data rate range is 1kbps to 10 mbps.
9. **Priority based data transmission:** During the time of disease if the condition is critical then the data sensor directly transmit patient condition to the medical server for immediate health care whenever any drastic change occurs.
10. **Real time data transmission:** It is basically used to transfer the digital range, signal and elevation coordinate from remote locally sensor to the user computer.

B. Components of BSN

1. **Sensors-** Sensor is a device which basically helps in detecting various conditions in our body according to the environmental changes such as temperature, heat, etc and helps to transfer data from one device to other in a wireless mode. For ex: - ECG sensors, Spo2 sensors, etc. [8]

2. Processor – Processor is a electronic logic circuit which helps to perform various operations in it. It helps to control all the hardware and software of the devices.
3. Transceiver – It is a device consisting of two components first is the transmitter and second is the receiver which when combined shares the common circuitry or a single housing.
4. Battery – It is a combination of one or more cell through which device is powered up according to flow of electrons using chemical reaction.

C. Applications of BSN

1. Monitoring – Monitoring refers to the checking of health status of the user which do their work hazardous nature i.e. hazardous chemicals and nuclear companies. So, using Body sensor the data can be collected and transmitted to remote diagnosis where it is analyzed by the doctor whenever it is needed.
2. Authentication – As we all know that the security is one of the important feature for storing the data, various body sensors are developed keeping this thing in the mind. Sensor such as imaging processors, decision making sensors are being developed in such a manner that if there is a change in the physiological and the behavioural conditions then that data is not shown to that user or hacker.
3. Gaming--To make the life of people more interesting and active, various companies use body sensor networks in their devices such as LED TV, Laptop, etc. Using this gamer, can imagine that they are playing at a real location and perform various body movements such as playing tennis, cricket, sword fighting, etc. One such example of gaming is PSP.
4. Performance Check--To keeps the record of the physio-logical condition such as blood pressure, heart rate, etc of the sport players. Body sensors are used so that after the analyzing the data the exercise routine and the desired performance goal can be achieved.
5. Motion Detection – [7] As we know elderly people mostly have a weak bone due to which they have a chance of falling and hurting them self. So, by using sensors user can easily be notified.

IV. WORKING OF SENSORS

Through this paper, we want to create a device through which the family member of any alzheimer patient can find out about their activity. [6] In this first the sensor would check the patient vital condition and then send their health report to a medical portal from where its report get converted into

database and is then send to the doctor for the analysis of the patient health.

For instance, the condition of the heart of the people can be checked by this type of algorithm. Through this algorithm, the sensing power of the patient can be detected. In this, the sensors are used in the body to check the movement of the legs by checking how much time, speed and vibration are covered by their walking. It also tells if the person is running, walking or going to get fall. For instance, if there is a change in motion of the day to day life of the patient then the family members can easily be denoted of it so that they can check about it and take precaution regarding their health

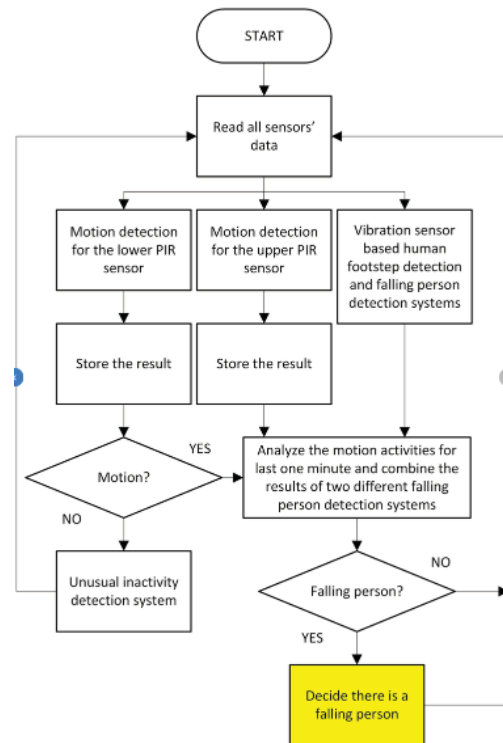


Fig 2 Algorithm which Shows Motion Sensing

V. IMPORTANCE OF WBSN IN TODAY'S WORLD

In today's world the life of people has become so hectic and hardworking that the people are not able to take care of themselves. One of the main roles of sensors is to make life of people happy go lucky. In order to maintain a very healthy life, wearable body sensor is must as they help to maintain our body in healthy mode and help us to lead our life peacefully and without any tension.

Sensor consist of various nodes whose work is to manage, process and analyze the whole condition of the patient's body and then give a report of it to the doctor and patient itself in the cloud. [5] Let us explain it through a particular example i.e. Smart watch. Smart watch refers to a digital watch whose major role is to keep update about the user's day to day activity and manage their heath by telling how much calories

or fat they have burnt while walking, jogging, etc. The most important role of the watch is to monitor the user heart rate, blood pressure and pulse rate. If due to some circumstance or excess work if there is a change in our body then that sensors immediately collect all the data and then analyze it, gives an warning reminder to notify the user about his/her health condition so that no major and fatal health issue occurs, thus saving his/her life by providing the correct information.

Secondly, the people especially elderly people who suffer from Alzheimer disease's (a disease which affects the brain so much that the retaining power of that person gets destroyed which leads the person to forget his family, his work and many more) can be helped with the help of this wireless sensing device. As the sensing device consist of GPS in it, so if by chance if any Alzheimer person or some other patient health gets deteriorate (such as if someone gets stroke, heart attack unexpectedly) then with the help of GPS and the sensing nodes the patient family as well the doctor who is in touch with that sensing network could track down his location and can save that patient in the nick of time.

VI. CONCLUSION

Therefore, we can say that, by using the sensing networks the patient or any other person can keep a check on its health conditions anytime and anywhere. Secondly through this sensors we are trying to make the life of people more easy who are suffering from the case of memory loss by thinking of attaching a image sensing node to that device which we would make so that whenever the user sees someone or anything for first time, it would create a image of it and the user would not face any difficulty during recognizing process. Finally we can conclude by saying that for making our body healthy and fit, the user need to keep check on medical condition by using

various body sensing technologies like Digital smart phone, etc.

ACKNOWLEDGMENT

We would like to express our sincere gratitude to our faculty mentor and Accendere Knowledge Management Pvt. Ltd for providing us the platform and opportunity to pursue the research

REFERENCES

- [1] Nabi Sultan, "Making use of cloud computing for healthcare provision: opportunities and challenges", International Journal of Information Management, University Campus Suffolk, United Kingdom, vol. 32, pp 177-184, April 2014
- [2] Mathew White, Girija Chetty, "Body Sensor Networks for Human Activity Recognition", IEEE, 2016 3rd International Conference on Signal Processing and Integrated Networks (SPIN), Noida, India, 2016
- [3] Abdelbaset Khalaf, Rishaad Abdoola, "Wireless Body Sensor Network and ECG Android Application for eHealth", IEEE, 2017 Fourth International Conference on Advances in Biomedical Engineering (ICABME), Beirut, Lebanon, 2017
- [4] Preeti, Kusum Grewal Dangi, Kumari Bharti Sangwan, "Wireless Body Area Sensor Networks", IEEE, 2017 International Conference on Computing and Communication Technologies for Smart Nation (IC3TSN), Gurgoan, India, 2017
- [5] Giancarlo Fotino, Mukaddin Pathan, Giuseppe Di Fatta, "BodyCloud: Integration of cloud computing and body sensor networks", IEEE, 4th IEEE International Conference on Cloud Computing Technology and Science Proceedings, Taipei, Taiwan, 2012
- [6] M. Pallikonda Rajasekaran, S. Radhakrishnan, P. Subbaraj, "Elderly patient monitoring system using a wireless sensor network", Telemedicine and e-health, vol 15, Feb 2009
- [7] Hande Alemdar, Cem Ersoy, "Wireless sensor networks for healthcare: a survey", Computer Networks, Istanbul, Turkey, vol 54, pp. 2688-2710, October 2010
- [8] Zhelong Wang, Cong Zhao, Sen Qui, "A human vital signs monitoring and activity recognition based on body sensor networks", Sensor Review, Vol 34, pp. 42-50, 2014